Bayesian Inference

Inferring hidden causes and values

- As we discussed before, we learn from experience about values in the world, that guide our decisions.
- We talked about reinforcement learning.
- But think about other inference problems – detecting one's intentions, the reason the floor is wet, that there is no line in Aroma, or why your friend is not texting.

Inference problems

- Usually there are many possible causes for an observation, and we try to think which one is more plausible, or probable.
- Can you give an example?
- What are the different possible causes?
- What is the probability of observation, given each of these causes?

Probability language

- Probability of an observation given a cause can be written:

$$p(Cause|Observation)$$

- Do you remember what it means?

Dependent probabilities

- In the Iraqi game of Mheibes, one of 45 men is hiding a ring in their arm. You need to guess where it is.
- Let's assume you narrowed it down to the last person.
- What is the probability of the ring being in his right hand? And left?
- $p(Right = Ring) =?$
- $p(Left = Ring) =?$



Anthony, J. (2025, June 14). *The World's Hardest Bluffing Game*. The Atlantic. https://www.theatlantic.com/magazine/archive/2025/07/mheibes-iraq-game/682901/

Dependent probabilities

- Now let's assume that the ring is in the left hand.
- This changes the probability of the ring being in the right hand, as these events are dependent.
- The ring cannot be in the left and in the right hand at the same time, or not in any of them.
- $p(Right = Ring|Left = Ring) =?$



Anthony, J. (2025, June 14). *The World's Hardest Bluffing Game*. The Atlantic. https://www.theatlantic.com/magazine/archive/2025/07/mheibes-iraq-game/682901/

Dependent probabilities

- $p(Right = Ring|Left = Ring) =?$

- This is different from independent probabilities, for example the probability of flipping a coin in the right hand and getting ‘H’, and flipping the coin with the left hand and getting ‘H’.
-
- $p(Right = H|Left = H) =?$



Anthony, J. (2025, June 14). *The World’s Hardest Bluffing Game*. The Atlantic. https://www.theatlantic.com/magazine/archive/2025/07/mheibes-iraq-game/682901/

	No	Ring
No	0	0.5
Ring	0.5	0

$$p(Right, Left) = p(Right|Left) \cdot p(Left)$$

	H	T
H	0.25	0.25
T	0.25	0.25

$$p(Right, Left) = p(Right|Left) \cdot p(Left)$$

$$p(Right, Left) = p(Right) \cdot p(Left)$$

Bayes Theorem

- $p(Right, Left) = p(Right|Left) \cdot p(Left)$
- $p(Left, Right) = p(Left|Right) \cdot p(Right)$

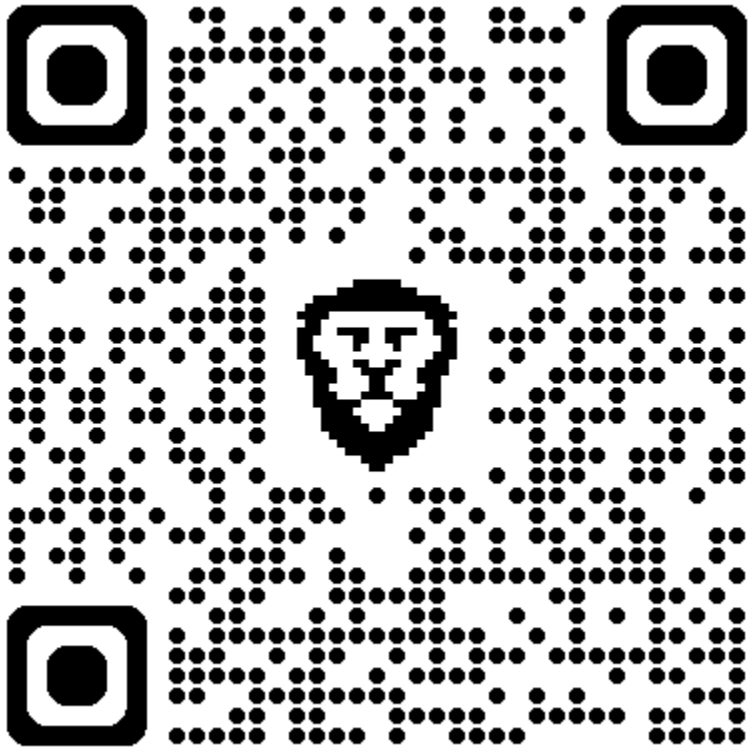
- $p(Right|Left) = \frac{p(Left|Right) \cdot p(Right)}{p(Left)}$



This is not the reverend Thomas Bayes (although it might be…)

Inference again…

- Let’s play coin flips (Task 1)
- https://sdl-exp.site/CompCogCourse/Lecture9Bayes/CoinFlipBayes.html



Bayes theorem and inference

- We wrote the inference problem as trying to solve:

$$p(Cause|Observation)$$

- With Bayes rule we can write it like this:

$$p(Cause|Observation) = \frac{p(Observation|Cause) \cdot p(Cause)}{p(Observation)}$$

- We will ignore the denominator …

Bayes theorem and inference

$$p(Cause|Observation) \propto p(Observation|Cause) \cdot p(Cause)$$

- We call the different parts of this equation in different names:

$$Posterior \propto Likelihood \cdot Prior$$

Bayes theorem and inference

$$p(Cause|Observation) \propto p(Observation|Cause) \cdot p(Cause)$$

$$Posterior \propto Likelihood \cdot Prior$$

- Prior – is what we currently think about the cause.
- Likelihood – the probability of current observation given the cause
- Posterior – our updated belief about the cause given the observation

Bayes theorem and inference

- We can infer the cause given a current set of observation and prior belief, or update our prior as we gather more observation.
- In this case, our posterior becomes the prior for the next observation.
- Let’s go back to our coin flipping game, and use the second task.
- https://sdl-exp.site/CompCogCourse/Lecture9Bayes/CoinFlipBayes.html

Coin flips

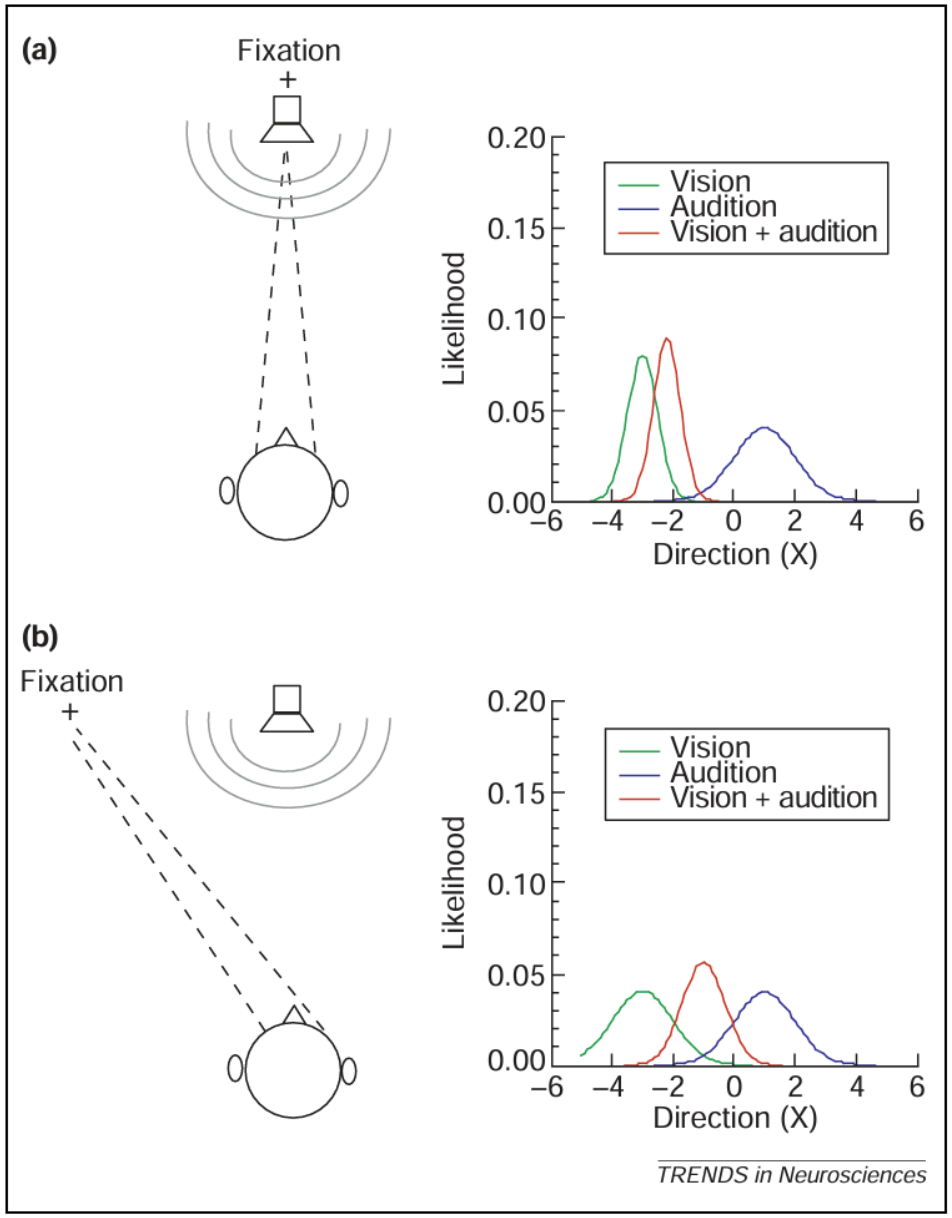
- What was the shape of the likelihood distribution?
- Did it sum up to 1? Why?
- What was the shape of the posterior?
- Did it sum up to 1?

Note that we already seen this form before…

- $Posterior \propto Likelihood \cdot Prior$
- $Q_{t+1} = Q_t + \alpha \cdot PE$
- $y_{t+1} = y_t + \epsilon_v$

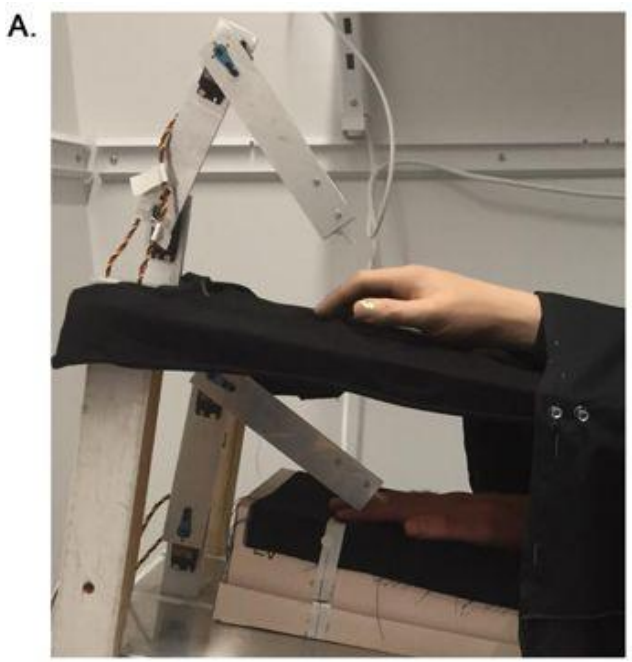
Bayesian inference in the brain

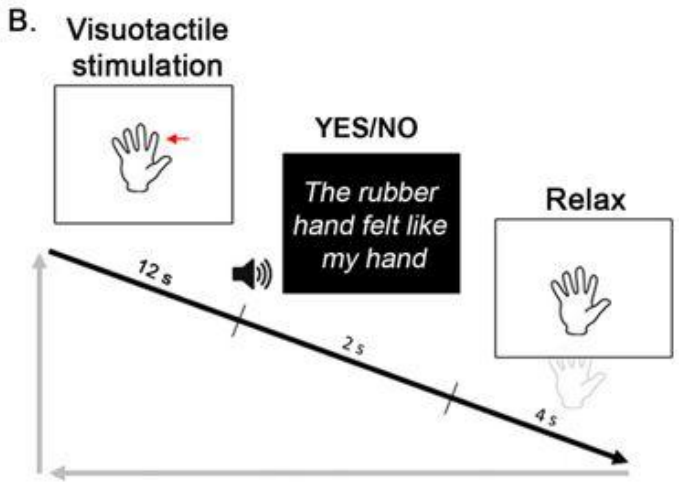
- Conflicting information between senses can be resolved using Bayesian inference.
- Likelihoods and priors are combined to form a posterior belief.

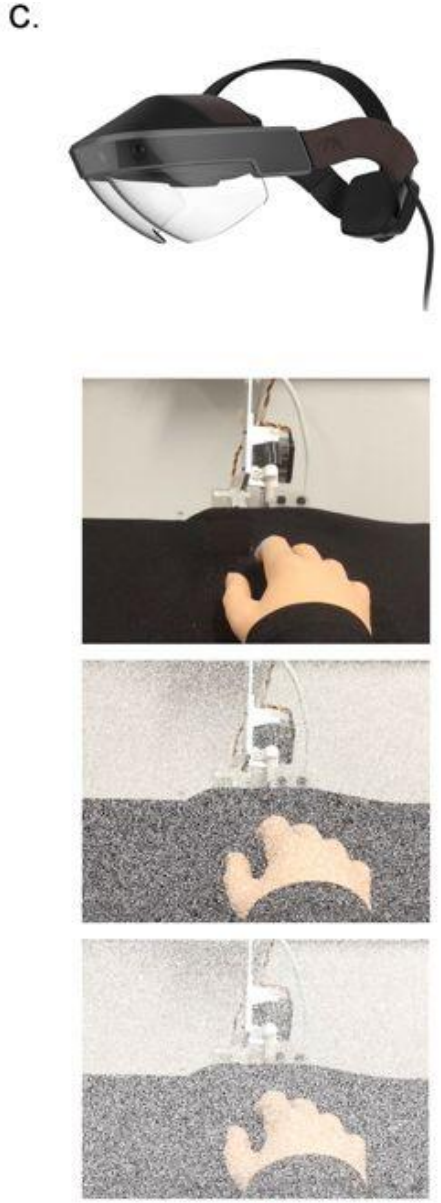


Knill, D. C., & Pouget, A. (2004). The Bayesian brain: the role of uncertainty in neural coding and computation. *Trends in Neurosciences*, *27*(12), 712–719.

Causal inference



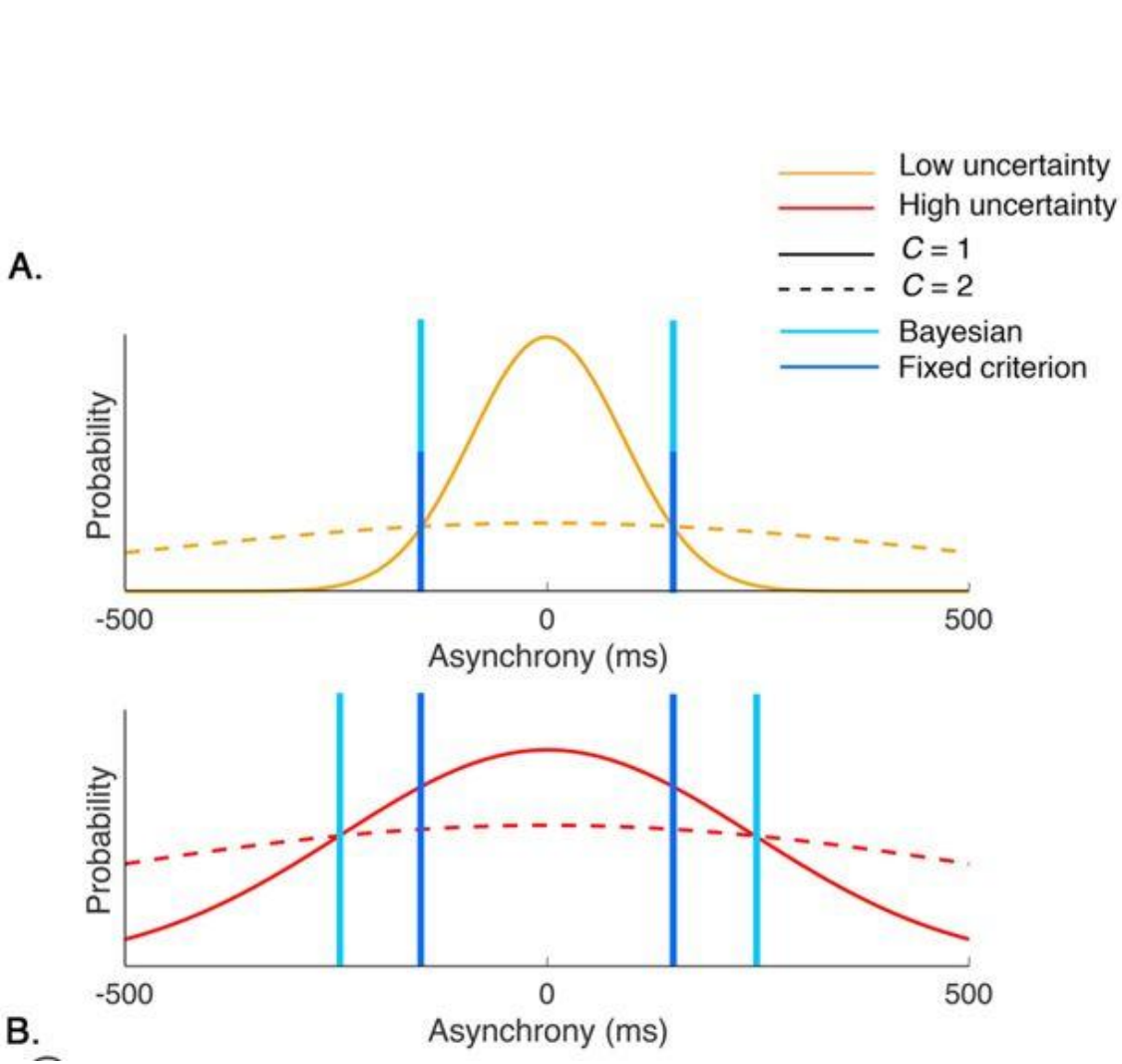


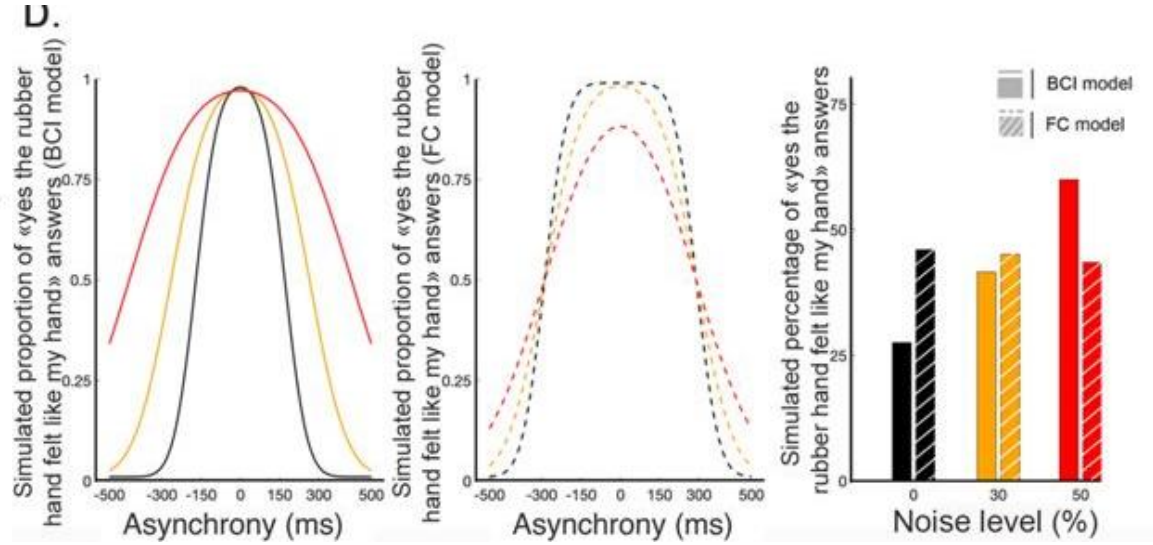


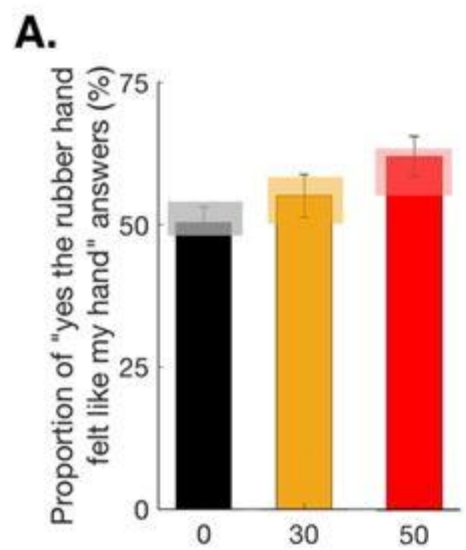
Chancel, M., Ehrsson, H. H., & Ma, W. J. (2022). Uncertainty-based inference of a common cause for body ownership. *eLife*, *11*. https://doi.org/10.7554/eLife.77221

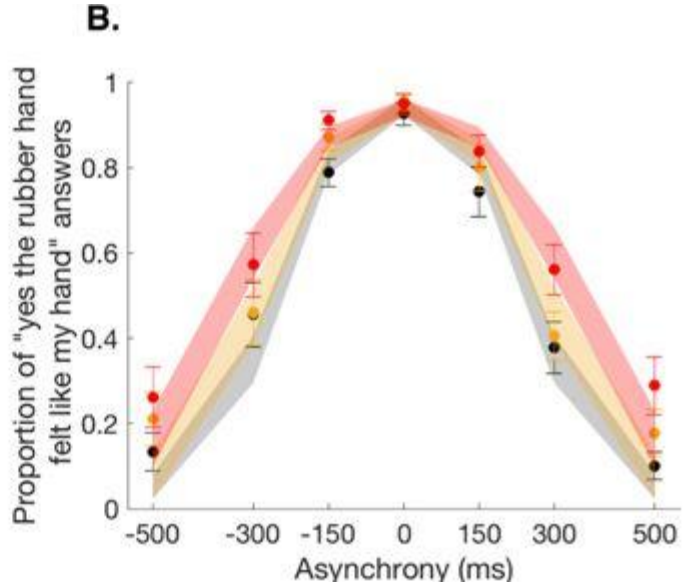
Choosing which cause is more plausible – same hand (C=1) or different (C=2)

- A posterior ratio was calculated:
- $d = log \frac{p(C=1|x)}{p(C=2|x)}$
- And rewritten as the sum of prior ratio and likelihood ratio:
- $d = log(\frac{psame}{1-psame}) + log(\frac{p(xtrial|C=1)}{p(xtrial|C=2)})$
- Which allowed the authors to determine the criterion for the limits of the rubber hand illusion.

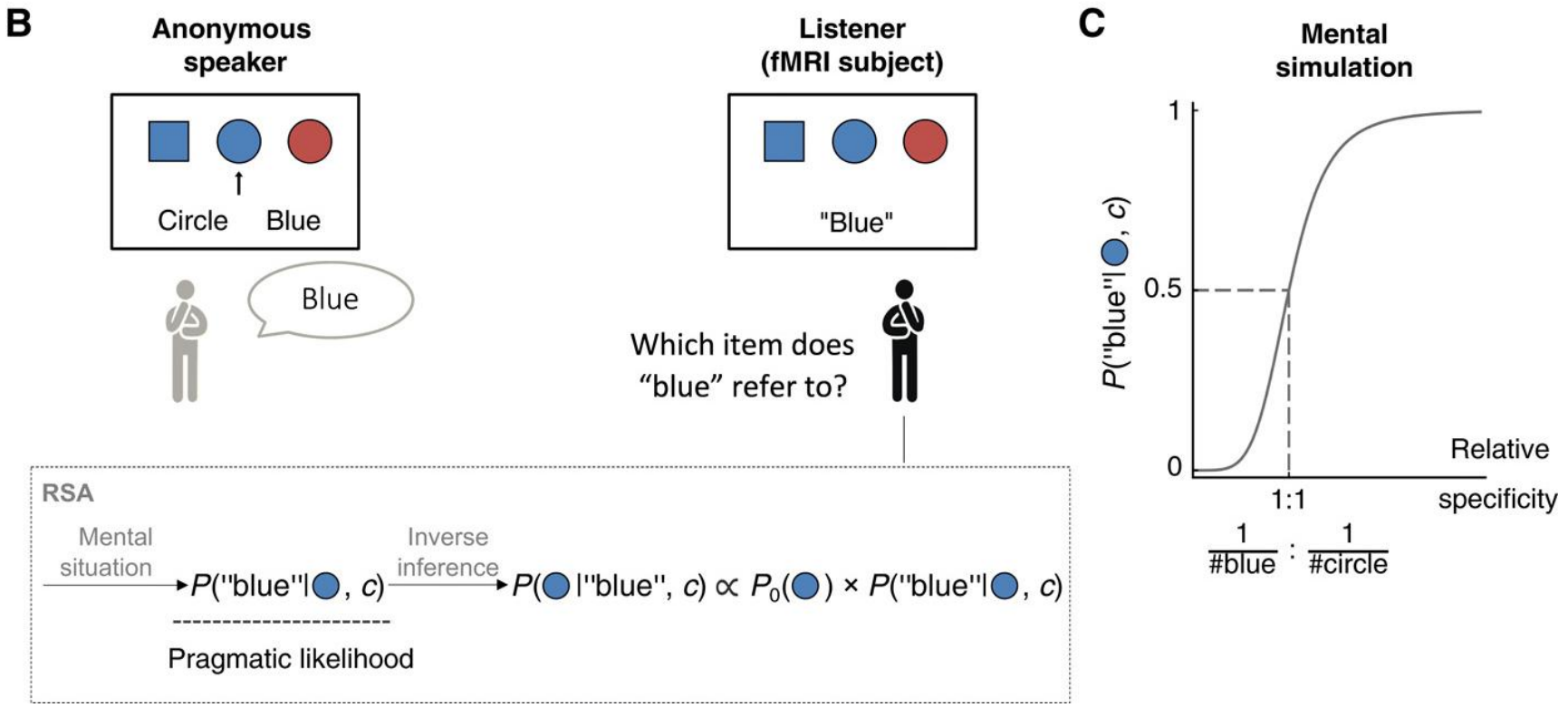

A.
Low uncertainty
High uncertainty
C = 1
C = 2
Bayesian
Fixed criterion
Probability
-500
0
500
Asynchrony (ms)
Probability
-500
0
500
Asynchrony (ms)
B.


D.
Simulated proportion of «yes the rubber hand felt like my hand» answers (BCI model)
Asynchrony (ms)
Simulated proportion of «yes the rubber hand felt like my hand» answers (FC model)
Asynchrony (ms)
Simulated percentage of «yes the rubber hand felt like my hand» answers
BCI model
FC model
Noise level (%)

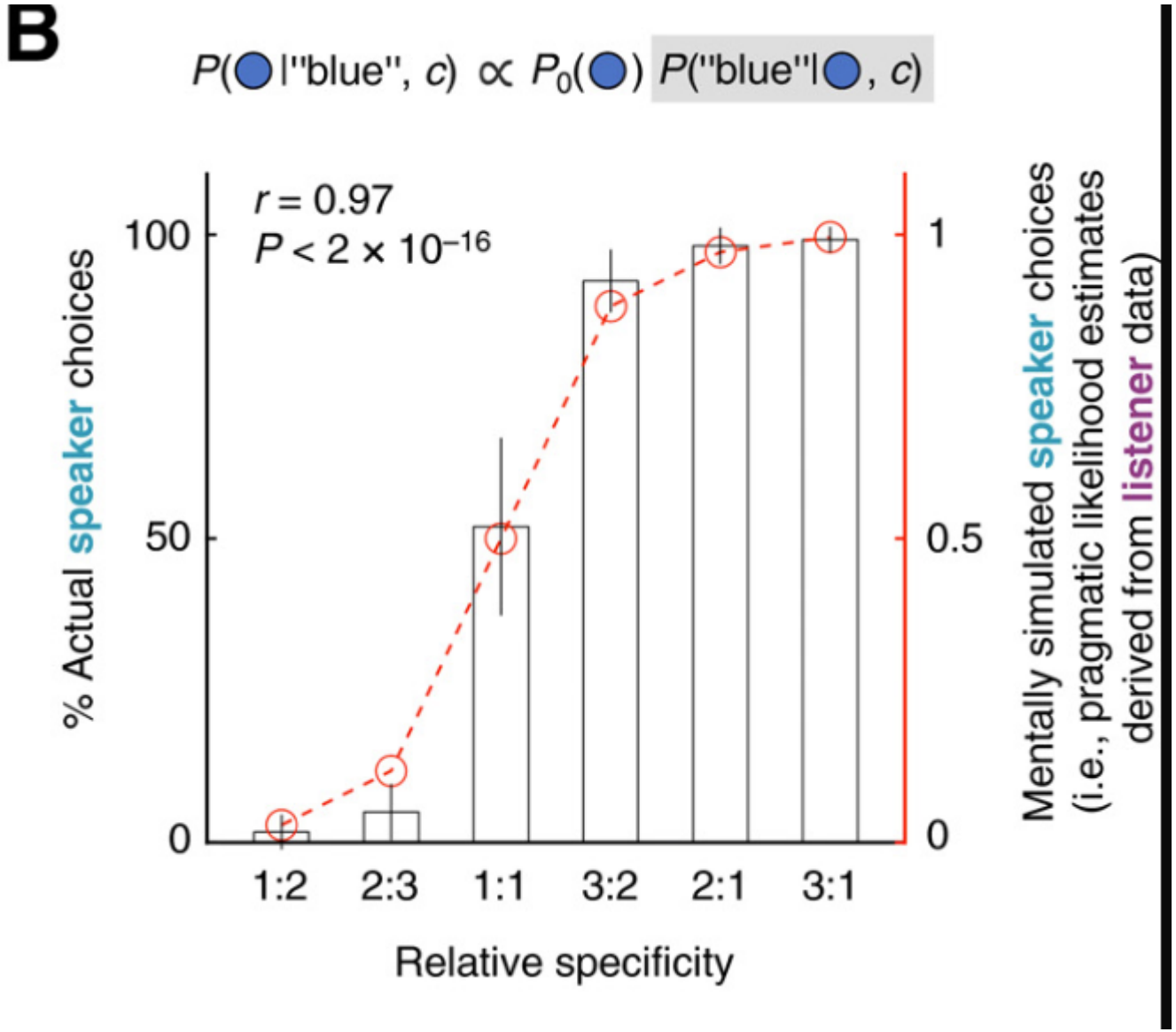

A.
Proportion of "yes the rubber hand felt like my hand" answers (%)
0
30
50


B.
Proportion of "yes the rubber hand felt like my hand" answers
Asynchrony (ms)

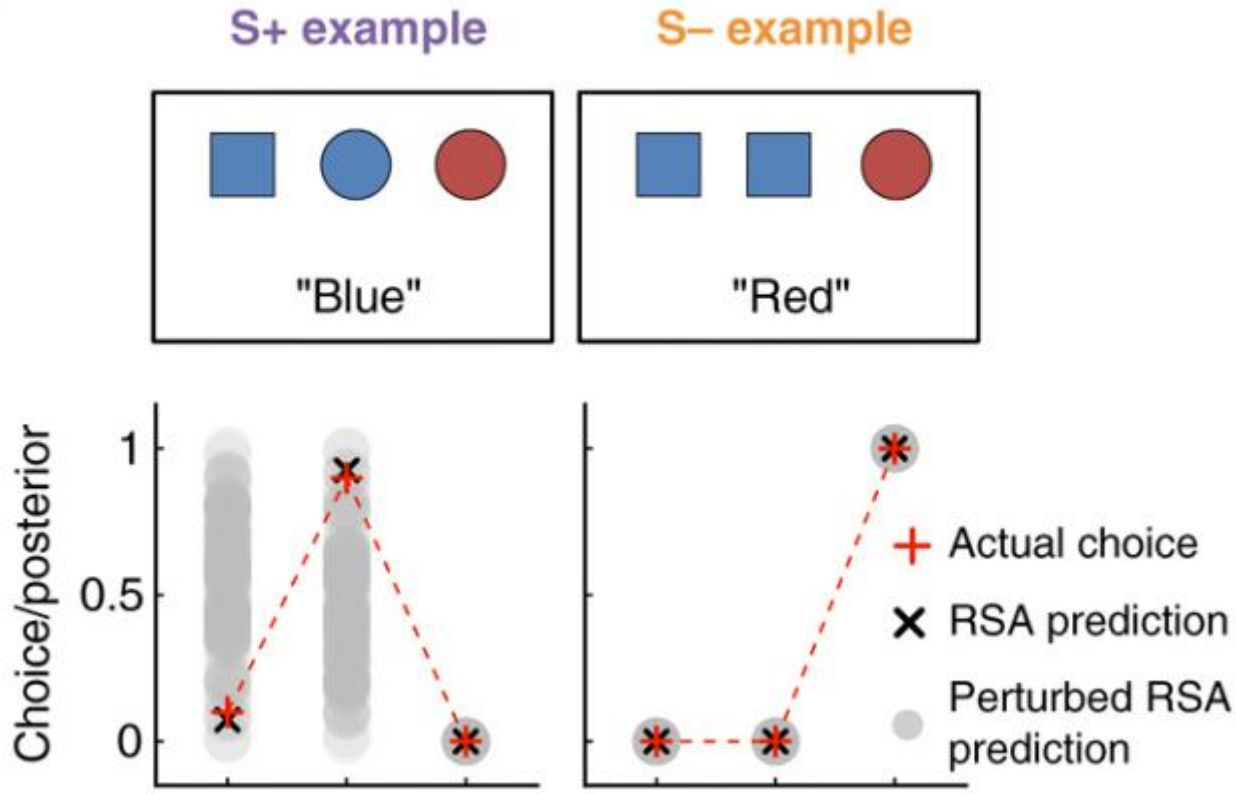
Inferring what other people think from what the say



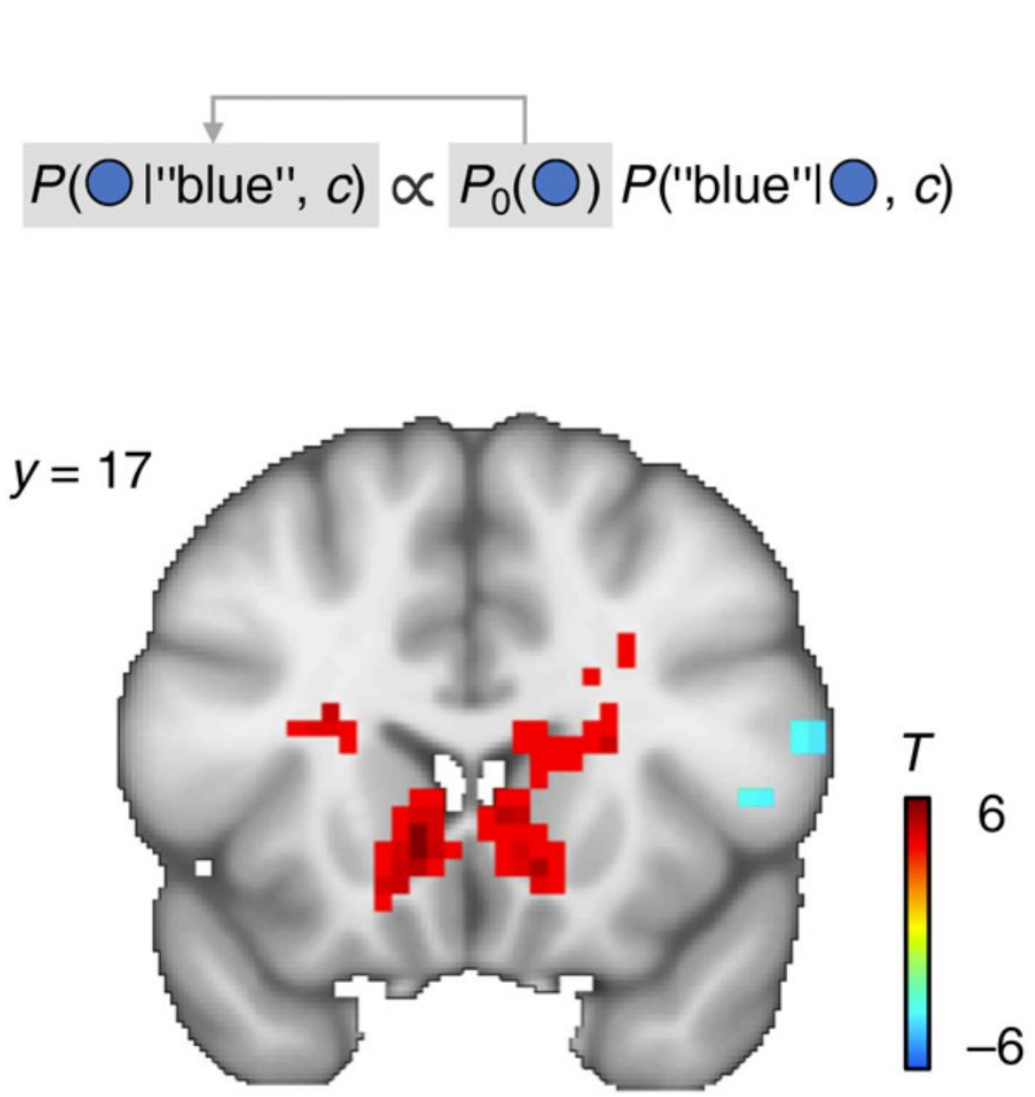
Mi, Q., Wang, C., Camerer, C. F., & Zhu, L. (2021). Reading between the lines: Listener's vmPFC simulates speaker cooperative choices in communication games. *Science Advances*, *7*(10), eabe6276.


B
P(● | "blue", c) ∝ P0(●) P("blue" | ●, c)
r = 0.97
P < 2 × 10−16
100
50
0
1
0.5
0
% Actual speaker choices
Mentally simulated speaker choices
(i.e., pragmatic likelihood estimates
derived from listener data)
1:2
2:3
1:1
3:2
2:1
3:1
Relative specificity

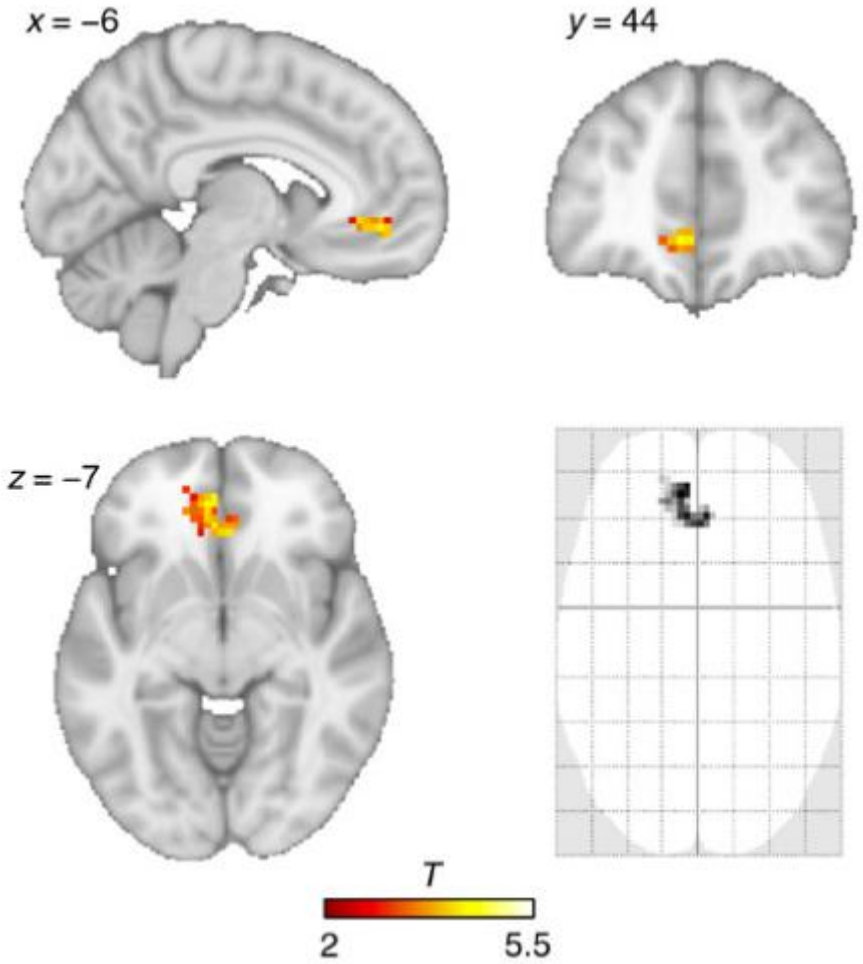
C


S+ example
S– example
"Blue"
"Red"
Choice/posterior
1
0.5
0
Actual choice
RSA prediction
Perturbed RSA prediction

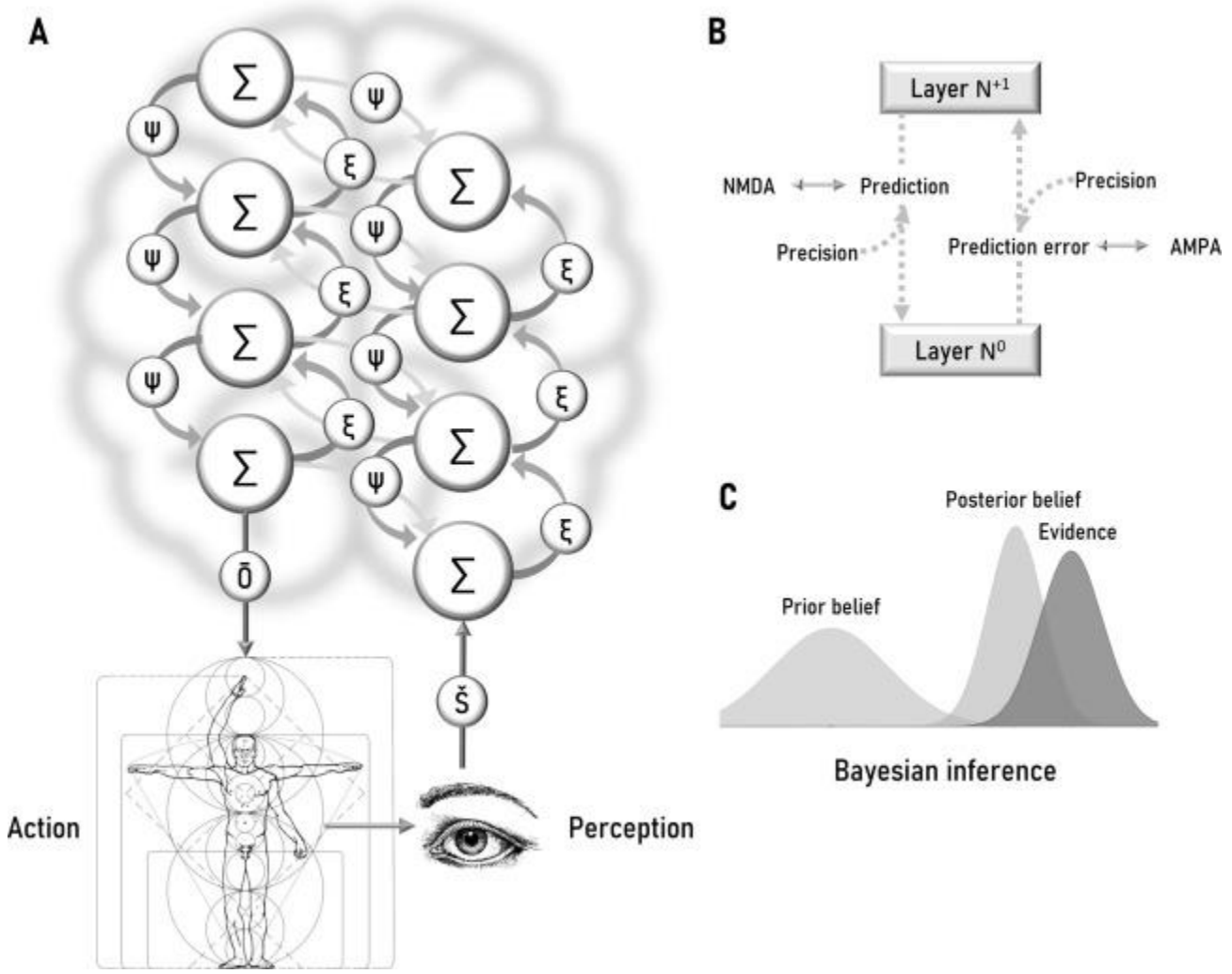
- **Listener striatum represents update-related signals.**
- (**A**) The listener bilateral striatum represents the update estimate (posterior-prior probability) for the chosen object at the time of expression onset.



- **Listener vmPFC encodes pragmatic likelihood estimates, above and beyond prior, posterior, and reaction time (RT).**
- BOLD activity in the listener vmPFC is significantly correlated with pragmatic likelihood (PL) estimates of the chosen object, controlling for prior, posterior, RT, and trial type (S+/S−)



Bayes inference as a model of the brain



- Hierarchical predictive processing in Bayesian brain theory. (A) Bayesian brain theory (BBT) supposes that the brain encodes a hierarchy of beliefs (∑) that are used to generate goal-directed actions (Ō) and predictions about sensory signals (Ψ) and sensory signal processing (Š). These predictions are updated by prediction errors (ξ) resulting from actions and changes in the environment. The predictive processing of sensory signals ensures the constant and dynamic adaptation of an organism to the evolution of its environment.

Bottemanne, H. (2025). Bayesian brain theory: Computational neuroscience of belief. *Neuroscience*, *566*, 198–204.

Likelihood can be a very useful metric

- It is often being described using its own letter, but in reverse order:

$$L(Cause|Observation) = p(Observation|Cause)$$

- And in modelling, it describes the relation between the data and our parameters estimations:

$$L(\theta|data) = p(data|\theta)$$

Let’s try to explain a series of observations.

- https://sdl-exp.site/CompCogCourse/Lecture9Bayes/MarblesDraw.html

Likelihood function

- The likelihood function can be used to answer – given what I just saw, how likely are different causes, in our case model parameters' values.
- We can create this likelihood function by simulations, like in the app, or analytically (in a second).
- What is our best guess of the real value of the parameters?
- MLE = maximum likelihood estimator

Bernoulli distribution function

- For getting 2 red out of 10 balls, we have a formula:



$$L(b) = p(D|\,\mathrm{b}) = \prod_{i=1}^{n} p(x_i|b) = \prod_{i=1}^{n} b^{x_i}\,(1-b)^{(1-x_i)}$$

Log-Likelihood – why?

- For getting 2 red out of 10 balls, we have a formula:



$$L(b) = p(D|\,\mathrm{b}) = \prod_{i=1}^{n} p(x_i|b) = \prod_{i=1}^{n} b^{x_i}\,(1-b)^{(1-x_i)}$$

$$\ell(b) = \log b \sum_{i=1}^{n} x_i + \log(\,1-b) \sum_{i=1}^{n} (1-x_i)$$

Maximum likelihood estimator

- We can calculate the values of the likelihood and find the maximum likelihood value.

$$\hat{\theta} = \arg\max_{\theta} L(\theta \mid y) = \arg\max_{\theta} P(\mathrm{y} \mid \theta)$$

- We can also do it analytically – when the derivative is 0:

MLE for Bernoulli

- $\ell(b) = \log b \sum_{i=1}^{n} x_i + \log(1-b) \sum_{i=1}^{n} (1 - x_i)$
- $\ell(b) = \log b \, N_O + \log(1-b) N_B$
- $\frac{\partial l(b)}{\partial b} = \frac{1}{b} N_O - \frac{1}{1-b} N_B = 0$
- $b = ?$

Likelihoods are useful for model comparisons!

- Instead of other types of errors, many models provide us with a predicted probability (of choices, for example, in RL).
- We can use likelihood as the measure of fit – the higher it is, the better our fit!
- High likelihood means that our model assigned high probability to the participant's choice.
- Because we have many choices, and many probabilities, we usually calculate log-likelihood.

Comparing models – log-likelihood ratio

- We cab compare two models that are nested in each other using ratio test:
- $LR = -2log\frac{L_{Specific}}{L_{General}} = -2 * (\log L_{Specific} - logL_{General})$
- Why –log?
- When is it positive, negative?
- We can test if this difference is significant using χ^2 test, with the difference in the number of parameters as df.
- This works when one model (specific) is the extension of another model (general), i.e. if setting some parameters values can make them identical.

Remember variance-bias? And preference for simple models?

- Information criteria use likelihoods and the number of parameters to set a score for a model.
- You can the directly compare these scores between models.
- AIC – Akaike Information Criteria, penalizes the model according to the number of parameters K.

Akaike 1973

$$AIC = -2 \ln L(\theta \mid y, M) + 2K$$

- The lower the AIC score, the better (see the – sign).

Another information criterion

- BIC – Bayesian Information Criterion – is even more severe – it penalizes according to the number of parameters K, and observations N.

Schwartz 1978

$$BIC = -2 \ln L(\theta \mid y, M) + K \ln N$$

Other criterions..

- Other criterions penalize based on variance in fit (WAIC, DIC).
- All these criteria were invented to make sure that adding more parameters is done with caution.

